

Notes: De Loecker & Warzynski (AER, 2012)

Markups and Firm-Level Export Status

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1 Big picture

- **Question.** How can we estimate firm-level markups from standard production data, and what do those markups tell us about exporters versus nonexporters?
- **Core idea.** Under cost minimization, the output elasticity of a flexible input equals its revenue share times the markup. If we can estimate output elasticities from a production function, we can back out markups without specifying a full product-market demand system.
- **Main methodological contribution.** The paper combines Hall-style markup measurement with the control-function approach from the production-function literature, so markups can be estimated while controlling for unobserved productivity.
- **Main empirical setting.** Slovenian manufacturing firms, 1994–2000, during a period of rapid transition, major restructuring, high firm turnover, and sharply rising export participation.
- **Main results.** Markups estimated with productivity controls are materially higher than in the standard Hall/Klette approaches; exporters have higher markups than domestic firms; and markups rise when firms enter export markets.
- **Interpretation.** The familiar exporter “productivity premium” partly reflects markup differences, not just lower marginal costs. Exporters appear to charge higher prices relative to marginal cost, and this gap widens at export entry.
- **Broader implication.** Studies that infer productivity growth or gains from exporting from revenue data can overstate true efficiency gains if they ignore changing markups, especially when export participation is rising over time.

2 Data and measurement (key objects)

2.1 Slovenian manufacturing panel

- **Source.** Slovenian Central Statistical Office; full company accounts for an unbalanced panel of manufacturing establishments/firms.
- **Coverage.** 7,915 firms over 1994–2000.
- **Observed variables.** Output/revenue, labor, capital, materials, export status, export sales, entry, and exit.
- **Why Slovenia is useful.** The country moved from central planning to market competition after independence. The period features large restructuring, heavy entry, export expansion, and measured productivity growth.

2.2 Exporting and industrial dynamics

- **Firm turnover.** Annual exit rates are around 3%; entry rates are especially high at the beginning of the sample.
- **Export expansion.** The number of exporters rises from 1,738 in 1995 to 2,335 in 2000, while total manufacturing exports increase by about 42% over the sample period.

- **Labor productivity growth.** Labor productivity rises from 14.71 thousand tolars in 1995 to 21.26 thousand tolars in 2000.
- **Interpretation.** Slovenia is exactly the kind of environment where one expects strong selection, nontrivial productivity dynamics, and potentially large markup differences between firms.

2.3 Output and prices

- **Ideal object.** The methodology would ideally use physical output, because markups should be estimated from quantity-based production functions.
- **In practice.** The paper uses deflated sales as output and corrects the measured expenditure share using the estimated residual from the production-function first stage.
- **Implication.** Unobserved prices may bias the *level* of markups, but under the paper's logic they should not overturn the estimated relationship between markups and export status when the empirical object is a log-markup differential.

2.4 Inputs and markup measurement

- **Flexible input requirement.** The method needs at least one variable input without adjustment costs. In the value-added setup this is labor; in the gross-output setup the authors use materials as a robustness check.
- **Dynamic input.** Capital is treated as dynamic, so the method avoids having to impose a static first-order condition on capital or measure a user cost of capital.
- **Key empirical object.** Firm-specific output elasticities of variable inputs, estimated from production functions and then combined with observed expenditure shares.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production technology and cost minimization

A firm produces according to

$$Q_{it} = Q(X_{it}^1, \dots, X_{it}^V, K_{it}, \omega_{it}). \quad (1)$$

- **Interpretation.** Output depends on variable inputs, capital, and firm productivity.
- **Important assumption.** Capital can be dynamic, but at least one variable input must be freely adjustable.

The associated cost-minimization problem is written as

$$\mathcal{L}(X_{it}^1, \dots, X_{it}^V, K_{it}, \lambda_{it}) = \sum_{v=1}^V P_{it}^{X_v} X_{it}^v + r_{it} K_{it} + \lambda_{it} (Q_{it} - Q(\cdot)). \quad (2)$$

For any flexible input X_{it}^v ,

$$\frac{\partial \mathcal{L}_{it}}{\partial X_{it}^v} = P_{it}^{X_v} - \lambda_{it} \frac{\partial Q(\cdot)}{\partial X_{it}^v} = 0. \quad (3)$$

Multiplying by X_{it}^v/Q_{it} gives

$$\frac{\partial Q(\cdot)}{\partial X_{it}^v} \frac{X_{it}^v}{Q_{it}} = \frac{1}{\lambda_{it}} \frac{P_{it}^{X_v} X_{it}^v}{Q_{it}}. \quad (4)$$

- **Interpretation.** The output elasticity of a flexible input equals the input's spending share in output, scaled by the inverse of marginal cost.

Define the markup as price over marginal cost,

$$\mu_{it} \equiv \frac{P_{it}}{\lambda_{it}}. \quad (5)$$

Then the key identity becomes

$$\theta_{it}^X = \mu_{it} \frac{P_{it}^X X_{it}}{P_{it} Q_{it}}, \quad (6)$$

or equivalently

$$\mu_{it} = \theta_{it}^X \left(\alpha_{it}^X \right)^{-1}, \quad (7)$$

where α_{it}^X is the expenditure share of input X in total sales.

- **This is the paper's fundamental formula.** Estimate the output elasticity θ_{it}^X , observe the revenue share α_{it}^X , and recover the markup.
- **Why it matters.** The method does not require specifying Bertrand, Cournot, or monopolistic-competition demand. All of that is pushed into the reduced-form object μ_{it} .

3.2 Production-function setup

The paper restricts attention to a Hicks-neutral productivity process with common technology parameters:

$$Q_{it} = F(X_{it}^1, \dots, X_{it}^V, K_{it}; \beta) \exp(\omega_{it}). \quad (8)$$

In logs, the estimating equation is

$$y_{it} = f(x_{it}, k_{it}; \beta) + \omega_{it} + \epsilon_{it}. \quad (9)$$

- **Interpretation.** ϵ_{it} is unanticipated production noise and measurement error; ω_{it} is the productivity term firms observe when choosing inputs.
- **Core econometric problem.** Inputs are correlated with ω_{it} , so OLS is biased.

3.3 Materials proxy and first stage

The paper follows the OP/LP/ACF control-function tradition and uses materials demand as a proxy for productivity:

$$m_{it} = m_t(k_{it}, \omega_{it}, z_{it}), \quad (10)$$

which is inverted into

$$\omega_{it} = h_t(m_{it}, k_{it}, z_{it}). \quad (11)$$

- **Interpretation.** z_{it} collects other state variables that shift input demand, especially export status and input prices.
- **Why materials.** Material demand is monotone in productivity under a wide range of imperfect-competition models, so it can be used to proxy for otherwise unobserved productivity.

Under a translog value-added specification, output is

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_{ll} l_{it}^2 + \beta_{kk} k_{it}^2 + \beta_{lk} l_{it} k_{it} + \omega_{it} + \epsilon_{it}. \quad (12)$$

The first-stage regression is

$$y_{it} = \phi_t(l_{it}, k_{it}, m_{it}, z_{it}) + \epsilon_{it}, \quad (13)$$

where expected output is

$$\phi_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_{ll} l_{it}^2 + \beta_{kk} k_{it}^2 + \beta_{lk} l_{it} k_{it} + h_t(m_{it}, k_{it}, z_{it}). \quad (14)$$

- **Interpretation.** The first stage nonparametrically partials out productivity using materials demand.

3.4 Productivity law of motion and second stage

Productivity evolves as

$$\omega_{it} = g_t(\omega_{i,t-1}) + \xi_{it}, \quad (15)$$

where the paper allows observable firm choices such as export status to affect expected future productivity in extended specifications.

After the first stage, for any candidate β , productivity is recovered as

$$\omega_{it}(\beta) = \hat{\phi}_{it} - \beta_l l_{it} - \beta_k k_{it} - \beta_{ll} l_{it}^2 - \beta_{kk} k_{it}^2 - \beta_{lk} l_{it} k_{it}. \quad (16)$$

The production-function coefficients are then pinned down using moment conditions of the form

$$\mathbb{E} \left[\xi_{it}(\beta) \begin{pmatrix} l_{i,t-1} \\ k_{it} \\ l_{i,t-1}^2 \\ k_{it}^2 \\ l_{i,t-1}k_{it} \end{pmatrix} \right] = 0. \quad (17)$$

- **Interpretation.** Capital is predetermined, and lagged labor is used as an instrument because current labor responds to productivity.
- **Empirical implication.** This is what lets the authors estimate firm-specific output elasticities while controlling for endogenous input choice.

3.5 Output elasticities and corrected expenditure shares

For a translog value-added production function, labor's output elasticity is

$$\hat{\theta}_{it}^L = \hat{\beta}_l + 2\hat{\beta}_{ll}l_{it} + \hat{\beta}_{lk}k_{it}. \quad (18)$$

If output is observed with multiplicative error, the corrected input expenditure share is

$$\tilde{\alpha}_{it}^X = \frac{P_{it}^X X_{it}}{P_{it} \tilde{Q}_{it} / \exp(\hat{\epsilon}_{it})}. \quad (19)$$

- **Interpretation.** The first-stage residual is used to purge the observed revenue share of output noise unrelated to true demand or productivity.
- **Final markup estimate.** Plugging $\hat{\theta}_{it}^X$ and $\tilde{\alpha}_{it}^X$ into equation (6) yields firm-level markups.

3.6 Special case: Hall/Klette constant-markup framework

If markups are constant across firms and time, the Hall-style estimating equation becomes

$$\Delta y_{it} = \mu \Delta x_{it} + \Delta \tilde{\epsilon}_{it}, \quad (20)$$

where $\Delta x_{it} = \alpha_L \Delta l_{it} + \alpha_K \Delta k_{it}$ in the simplest Cobb-Douglas case.

The paper also derives a level version,

$$y_{it} = \mu l_{it}^* + \beta_k k_{it} + \omega_{it} + \epsilon_{it}, \quad (21)$$

with $l_{it}^* = \alpha_L^L l_{it}$, and a first-difference control-function version,

$$\Delta y_{it} = \mu \Delta l_{it}^* + \Delta \tilde{h}_t(m_{it}, k_{it}, z_{it}) + \Delta \epsilon_{it}. \quad (22)$$

- **Interpretation.** These special cases show exactly which assumptions the paper relaxes relative to the classic Hall and Klette approaches.
- **Main lesson.** Once productivity is controlled for properly, the estimated markup rises materially relative to Hall-style estimates.

3.7 Cross-sectional exporter markup regression

The first core regression is

$$\ln \mu_{it} = \delta_0 + \delta_1 e_{it} + b'_{it} \sigma + \nu_{it}, \quad (23)$$

where e_{it} is an export dummy and b_{it} includes labor, capital, and year-industry controls.

- **Interpretation.** δ_1 is the exporter markup premium in percentage terms.
- **Important caveat.** The authors do not interpret this cross-sectional coefficient as a causal parameter; it is a carefully measured conditional difference.

To separate markup differences from marginal-cost differences, they also estimate

$$(\ln P_{it} - \ln C_{it}) = \delta_0 + \delta_1 e_{it} + \delta_2 \omega_{it} + b'_{it} \sigma + \nu_{it}. \quad (24)$$

- **Interpretation.** Once productivity is included, the export dummy isolates price-side differences that are not just lower marginal costs.
- **Empirical result.** The exporter coefficient falls from about 0.076 to about 0.021, but remains positive; roughly 30% of the markup gap survives conditioning on productivity.

3.8 Export-entry regression

To study dynamics around export participation, the paper estimates

$$\ln \mu_{it} = \gamma_0 + \gamma_1 \text{Entry}_{it} + \gamma_2 \text{Exit}_{it} + \gamma_3 \text{Always}_i + b'_{it} \sigma + \nu_{it}. \quad (25)$$

- **Interpretation.** γ_1 measures the within-firm change in markup after export entry relative to the pre-entry period; γ_2 does the same for quitting export markets.
- **Empirical role.** This is the paper's dynamic evidence that markups rise after firms start exporting.

Productivity itself is recovered from the first-stage fitted value as

$$\hat{\omega}_{it} = \hat{\phi}_{it} - f(x_{it}, k_{it}; \hat{\beta}). \quad (26)$$

The authors then estimate additional correlations between markups and productivity, and they revisit the Hall productivity-growth decomposition using

$$\Delta y_{it} - \hat{\mu} \Delta \tilde{x}_{it} - \hat{\beta}_k \Delta k_{it} = \Delta \omega_{it}. \quad (27)$$

- **Interpretation.** Measured productivity growth is overstated when estimated markups are too low, and the bias becomes worse when exporters have systematically higher markups.

3.9 Robustness to unobserved prices

The paper emphasizes that with revenue data, the estimating object is effectively

$$(\ln \theta_{it}^X - \ln \alpha_{it}^X) = \theta_0 + \theta_1 e_{it} + \nu_{it}. \quad (28)$$

- **Interpretation.** Under Cobb-Douglas, any omitted-price bias mainly enters the constant term, so the exporter markup differential remains identified.

3.10 Exporter markups by destination market

If one could allocate inputs across markets, exporter markups by destination $s \in \{d, e\}$ would satisfy

$$\mu_{it}^s = \theta_{it}^L \left(\rho_{it}^s \frac{w_{it} L_{it}}{(P_{it} Q_{it})_s} \right)^{-1}, \quad (29)$$

where ρ_{it}^s is the share of the wage bill used for output sold on market s .

The average markup change at export entry can be decomposed as

$$\Delta \mu_{it} = \Delta(\rho_{it}^d \mu_{it}^d) + \rho_{it}^e \mu_{it}^e, \quad (30)$$

or, since entrants were purely domestic before entry,

$$\rho_{it}^d \mu_{it}^d + \rho_{it}^e \mu_{it}^e > \mu_{i,t-1}^d. \quad (31)$$

- **Interpretation.** The observed average markup increase after export entry could come from higher export-market markups, higher domestic markups, or both.
- **Authors' suggestive finding.** For firms with very small export shares at entry, the total average markup increase is negligible, suggesting domestic markups barely move and most of the action comes from export-market markups.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Key comparison.** The paper uses within-industry variation in flexible input use, expenditure shares, and productivity-controlled output elasticities to recover markups.
- **What is being compared?** Firms with different export status, and firms before versus after export entry, within a common accounting and production-function framework.
- **Main omitted-variable problem.** Productivity shocks move both inputs and output. If not controlled for, they bias output elasticities and thus bias markups.

4.2 Why the control-function approach is central

- **Materials proxy.** Materials demand is used to invert for firm productivity, following the OP/LP/ACF logic.

- **Predetermined inputs.** Capital and lagged labor identify production-function coefficients through moment conditions on the innovation to productivity.
- **Exporting as a state variable.** Export status is explicitly included in the control function and can also enter the productivity law of motion, so exporting is not treated as orthogonal to firm dynamics.

4.3 Why this is more credible than Hall/Klette for this question

- **Hall/Klette restriction.** Those approaches effectively impose a common markup and typically work in first differences, amplifying measurement error and leaving productivity-control issues unresolved.
- **This paper’s gain.** Markups are firm and time specific, so they can be related directly to export status, productivity, and entry dynamics.
- **Validation exercise.** When the paper imposes a common markup but uses the control-function logic, estimated markups still rise relative to Hall; this shows that productivity control is quantitatively important even before allowing for heterogeneous markups.

4.4 Assumptions users of the estimates should keep in mind

- Cost minimization and at least one flexible input.
- Hicks-neutral productivity and common technology parameters within industries.
- Monotone materials demand in productivity, conditional on capital and other state variables.
- Input prices are serially correlated enough that lagged labor or materials can instrument current flexible-input use.
- Unobserved prices may bias levels, but not the main log-markup differentials the paper studies.

5 Tables and figures

5.1 Table 1: Firm turnover and exporting in Slovenian manufacturing

Read:

- The Slovenian manufacturing sector is highly dynamic: exit is around 3% per year, entry is high, and the exporter count grows sharply.
- Labor productivity rises strongly over the same period, suggesting that export expansion and restructuring coincide with broad reallocation and efficiency gains.
- This table motivates why a markup-based reinterpretation of “productivity gains from exporting” is important here.

5.2 Table 2: Estimated markups across methods

Read:

Year	No. firms	Exit rate	Entry rate	No. exporters	Labor productivity
1995	3,820	3.32	13.14	1,738	14.71
1996	4,152	2.60	5.44	1,901	16.45
1997	4,339	3.43	4.47	1,906	18.22
1998	4,447	3.94	4.14	2,003	18.81
1999	4,695	3.26	3.30	2,192	21.02
2000	4,906	2.69	3.38	2,335	21.26

Note: Labor productivity is in thousands of tolar (deflated by industry-specific producer price index).

Methodology	Markup
Hall ^a	1.03 (0.004)
Klette ^a	1.12 (0.020)
<i>Specification</i>	
I (Cobb-Douglas)	1.17
II (I w/ endog. productivity)	1.10
III (I w/ additional moments)	1.23
IV (Translog)	1.28
V (II w/ export input)	1.23
VI (Gross Output: labor)	1.26
VI (Gross Output: materials)	1.22
VII ^a (I w/ single markup)	1.16 (0.006)
VIII ^a (First difference)	1.11 (0.007)

^aMarkups are estimated jointly with the production function (as discussed in Section III), and we report the standard errors in parentheses. The standard deviation around the markups in specifications **I–VI** is about 0.5.

- Hall gives an average markup of about 1.03 and Klette about 1.12.
- The paper’s preferred specifications produce median markups in the 1.17–1.28 range, depending on value-added versus gross-output, Cobb-Douglas versus translog, and whether export status enters the productivity process.
- Two robustness messages stand out: using materials in gross output gives markups similar to labor-based measures, and imposing a static first-order condition on capital yields much larger markups (around 1.5–1.6), consistent with adjustment costs contaminating the capital wedge.

5.3 Table 3: Markups and export status, cross section

Read:

- Exporters charge higher markups than domestic firms in all preferred specifications.
- The exporter premium in levels is about 0.13–0.18, while Hall and Klette fail to detect a statistically meaningful premium.
- This is the paper’s central cross-sectional result: revenue-based exporter productivity premia partly reflect higher markups.

Methodology	Export Premium
Hall	0.0155 (0.010)
Klette	0.0500 (0.090)
<i>Specification</i>	
I (Cobb-Douglas)	0.1633 (0.017)
II (I w/ endog. productivity)	0.1608 (0.017)
IV (Translog)	0.1304 (0.014)
V (II w/ export input)	0.1829 (0.017)
VIII (First difference)	0.1263 (0.013)

Notes: Estimates are obtained after running equation (21) where the different specifications refer to the different markup estimates, and we convert the percentage markup difference into levels as discussed above. The standard errors under specifications **I–V** are obtained from a nonlinear combination of the relevant parameter estimates. All regressions include labor, capital, and full year and industry dummies as controls. Standard errors are in parentheses.

Method output elasticity	Export entry effect	
	Percentage (γ_l)	Level (μ_w)
I (Cobb-Douglas)	0.0467 (0.0127)	0.0939 (0.0260)
II (I w/ endog. productivity)	0.0467 (0.0127)	0.0925 (0.0250)
IV (Translog)	0.0481 (0.0128)	0.0797 (0.021)
V (II w/ export input)	0.0497 (0.0127)	0.0994 (0.0260)
VIII (First difference)	NA	0.0700 (0.022)

Notes: The standard errors under **I–V** are obtained from a nonlinear combination of the relevant parameter estimates. We drop the estimates from specifications **III** and **VI** since they are identical to the ones reported in this table. The latter is as expected since the estimate of the capital coefficient does not impact the markup estimates for instance. Specification **VIII** delivers an immediate estimate of the level impact on markups. All regressions include labor, capital, and full year and industry fixed effects as controls.

5.4 Table 4: Markups and export status, export entry effect

Read:

- Export entry raises markups by about 4.7–5.0% in the preferred specifications, or roughly 0.08–0.10 in levels.
- The first-difference restricted model still gives a positive effect, around 0.07 in levels, reinforcing the dynamic result.
- This table is the key time-series contribution of the paper: not only are exporters different, firms become higher-markup producers after entering export markets.

6 Short summary

- **Conclusion.** A production-function-based control-function approach can recover firm-level markups from standard data, and those markups are systematically higher for exporters and rise with export entry.
- **Intuition.** When firms enter export markets, they appear not only to become more productive but also to charge higher prices relative to marginal cost.
- **Empirical lesson.** Revenue-based productivity measures mix together efficiency and market power. The mix matters especially in trade settings with changing export participation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Methodological result.** The paper shows how to estimate firm-level markups using a flexible-input first-order condition plus productivity-controlled production-function estimates.
- **Quantitative result.** Correctly estimated markups are higher than Hall/Klette estimates, with preferred medians around 1.2 rather than 1.0–1.1.
- **Trade result 1.** Exporters have significantly higher markups than nonexporters.
- **Trade result 2.** Markups increase after export entry.
- **Interpretation.** Exporter premia in revenue productivity are not only about lower marginal costs; they also reflect higher markups.
- **Aggregate implication.** If exporters have higher markups and export participation rises, measured aggregate productivity growth is biased upward when markups are ignored.

7.2 Economic intuition

- More productive firms can survive tougher competition and often charge higher markups; exporters are selected from this part of the distribution.
- Export entry may also move firms into markets with different demand elasticities or quality premia, which lets them price further above marginal cost.
- Because markups enter revenue directly, any empirical design that uses revenue as output risks confusing market power with technological efficiency.

7.3 Takeaway

This paper is important because it bridges industrial organization, production-function estimation, and trade. The markup identity itself is simple, but the paper shows that getting the output elasticity right is everything. Once productivity is handled carefully, the exporter premium looks partly like a markup premium, and the usual interpretation of revenue productivity in trade data becomes much less clean.